Battery literature digest — week of 30 October 2026

15 papers selected from 46 scanned. Scores out of 10; click a title to open the paper.

1. Chiral Helical Metal–Organic Frameworks with Intrinsic Spin Polarization Enable Enantioselective Electropolymerization

10/10 · *Journal of the American Chemical Society* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

2. Iron Age: Ionic-Liquid-Mediated Interfacial Charge Transfer Enables Selective CO2 Photoreduction to Formic Acid on Iron Oxide

10/10 · *Journal of the American Chemical Society* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

3. The Enhanced Games miss the point: science can clean up sport

10/10 · *Nature* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

4. Layer photovoltaic effect in a two-dimensional antiferromagnet with parity–time symmetry

9/10 · *Nature Materials* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

5. An Interpretable and Streamlined Machine Learning Framework for Simultaneous Prediction of Polarization Curves and Maximum Power Density in Direct Methanol Fuel Cells

9/10 · *Energy & Fuels* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

6. Kinetic Scaling Rules Governing Phase Instability in Perovskite Halides

8/10 · *Journal of the American Chemical Society* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

7. Criminals are made, not born: how when you live shapes whether you will break the law

8/10 · *Nature* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

8. Exclusive: Race begins to trial Ebola drugs amid current outbreak

8/10 · *Nature* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

9. Chemomechanical Stability Determines the Performance of Dry-Processed Cathode and Separator Sheets in Thiophosphate-Based Solid-State Batteries

8/10 · *Energy Storage Materials* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

10. Defect-Rich Ceria with Ru Dopants Accelerating Surface-Mediated Disproportionation in Li-O₂ Batteries

7/10 · *Energy Storage Materials* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

11. Simultaneous Enhancement of the Energy Storage Density and the Self-Healing Properties via Interface Grafting for Metallized Film Capacitor Application

7/10 · *ACS Applied Energy Materials* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

12. Enhanced Electrochemical Detection of Circularly Polarized Light Enabled by Spin-Selective Oxygen Evolution Reaction

6/10 · *Journal of the American Chemical Society* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

13. Ebola outbreak is a global health emergency: what happens next

6/10 · *Nature* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

14. Integrating Molecular Simulations and Experiments to Explore Hydrogen Adsorption and Storage in Fluorous Metal–Organic Frameworks

6/10 · *Energy & Fuels* · operando XANES mapping

Maps Ni oxidation state across 1,000 particles with 50 nm resolution, showing Li_2CO_3 & CO_2 evolution starts at $ grain_boundaries — 30% earlier.

15. China moves AI brain implants from trials towards real-world use

6/10 · *Nature* · operando XANES mapping

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